

Problem-Solution Fit Canvas

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Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

1. CUSTOMER SEGMENT(S) [CS]

- Government Policy Makers
 - Researchers and Data Analysts
 - International Development Organizations
 - Educational Institutions
 - Economists and Social Scientists
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2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Difficulty in predicting HDI accurately.
 - Manual analysis of development indicators is time-consuming.
 - Complex relationships between health, education, and income.
 - Lack of real-time analytical tools.
 - Frequent need for country comparison and development assessment.
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3. TRIGGERS TO ACT [TR]

- Need to evaluate a country's development level.
 - Requirement for evidence-based policy decisions.
 - Annual publication of HDI reports.
 - Research and comparative analysis of countries.
 - Development planning by governments and NGOs.
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4. EMOTIONS BEFORE / AFTER [EM]

Before

- Confused
- Uncertain
- Frustrated
- Concerned

After

- Confident
 - Informed
 - Satisfied
 - Motivated
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5. AVAILABLE SOLUTIONS – PROS & CONS [AS]

Existing Solutions

- Manual statistical analysis
- Excel-based calculations
- UNDP HDI reports
- Data visualization tools

Pros

- Reliable historical data
- Easy to access public datasets
- Standardized HDI methodology

Cons

- No predictive capability
 - Time-consuming analysis
 - Limited real-time decision support
 - Requires technical expertise
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6. CUSTOMER LIMITATIONS [CL]

- Limited technical knowledge
- Budget constraints
- Incomplete or outdated datasets
- Limited computing resources
- Internet connectivity issues

7. BEHAVIOR + ITS INTENSITY [BE]

- Frequently analyzes country development data.
- Regularly compares HDI rankings.
- Uses statistical reports for planning.
- High dependence on accurate predictions.

Intensity: High

8. CHANNELS OF BEHAVIOR [CH]

Online

- Government Portals
- UNDP Website
- Research Databases
- Web Application
- Interactive Dashboards

Offline

- Government Reports
 - Research Papers
 - Conferences
 - Policy Meetings
 - Educational Institutions
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9. PROBLEM ROOT / CAUSE [RC]

- Multiple indicators influence HDI simultaneously.
 - Manual calculations are slow and error-prone.
 - Large datasets are difficult to analyze.
 - Lack of intelligent prediction systems.
 - Limited visualization and analytical support.
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10. YOUR SOLUTION [SL]

Develop a **Machine Learning-based Human Development Index (HDI) Prediction System** that predicts HDI categories using **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita**. The system provides real-time predictions, interactive dashboards, country comparison, trend analysis, and downloadable reports to support policymakers, researchers, educators, and development organizations in making informed decisions.

Figure 6.1: Problem–Solution Fit Canvas for the Human Development Index (HDI) Prediction System

